Mohana Siddhartha Chivukula

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EDUCATION

Master of Science Computer Science Aug 2024 – May 2026
Arizona State University, Tempe, AZ 4/4 GPA
Relevant Coursework – Data Processing at Scale, Data Mining, Statistical Machine Learning, Mobile Computing

Bachelor of Technology Computer Science and Engineering Aug 2020 – May 2024
Shiv Nadar University, Delhi NCR, India 7.61/10 GPA
Relevant Coursework – Data Structures and Algorithms, Operating Systems, Information Retrieval, Big Data Processing

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, Dart, Kotlin, Go
Frameworks & Libraries: React, Next.js, Node.js, Flutter, Flask, TensorFlow, Scikit-learn, Pandas
AI & Machine Learning: LLM Integration, RAG, Generative AI, Computer Vision, Natural Language Processing, AWS Bedrock, Vector Databases, LangChain, LSTM, ARIMA, XGBoost, Random Forest, Logistic Regression, Vision Transformers
Infrastructure & Databases: Docker, Kubernetes, Kafka, AWS, CI/CD, Git, PostgreSQL, MySQL, MongoDB, Neo4j, Snowflake

PROFESSIONAL EXPERIENCE

Get SuperStars Inc., Remote: Software Developer Intern May 2025 – Aug 2025

- Reduced heap memory by **45%** through allocation profiling and widget tree refactoring in video-heavy components using **Dart DevTools** on a live cross-platform iOS and Android application.
- Eliminated redundant API calls by **50%** using Flutter and Dart, consolidating notification endpoints into a single stream-based polling cycle while integrating **8+** REST API endpoints with token-based authentication.
- Led a team of **3 interns** to deliver **4 production features** in **3 months**, becoming the **#1** contributor in the repository.

ZS Associates, Gurugram, India: Business Technology Solutions Associate Intern Jan 2024 – Jul 2024

- Led end-to-end development of distributed ETL pipelines using Snowflake and IICS consolidating **70+** vendor datasets into **4** data warehouse schemas and processing **10M+** records to deliver analytics ready data.
- Integrated **GPT-4** into vendor schema analysis, reducing the analysis time by **40%** and uncovering **100+** data anomalies that directly influenced architecture decisions for the DMT layer.
- "**Dashing Debut**" Award, ZS Associates H1'24 Rewards & Recognition.

ACADEMIC & OPEN SOURCE

Open-Source Contributor, LiteLLM Mar 2026 – Apr 2026

- Patched critical model routing failures in LiteLLM causing error tracebacks on every request to OpenRouter based deployments, restoring reliable fallback routing across production AI systems.

SCAI Grader, Arizona State University, School of Computing and Augmented Intelligence Aug 2025 – May 2026

- Mentored **100+** students across 3 courses, delivered technical lectures on Python and Android Development and building Automated Grading tools at Arizona State University.

PROJECTS

Promptly, (React, Next.js, TypeScript, Prisma, PostgreSQL, AWS, LangChain) Dec 2025 – Feb 2026

- Developed a multi-model AI platform using Next.js, TypeScript, and **AWS Bedrock**, querying **14 LLMs** in parallel.
- Implemented **RAG** pipeline using **LangChain** and pgvector for document-based context querying across all models.
- Built an **Agentic AI** pipeline using Claude Sonnet to classify model outputs by agreement and confidence score.

STREAM, (Python, Kafka, Kubernetes, Neo4j, Docker) Oct 2025 – Nov 2025

- Built a real-time **distributed streaming pipeline** using Kafka, Kubernetes and Neo4j, ingesting NYC Taxi data to enable graph analytics across **42 nodes** and **1530 relationships** using PageRank and BFS algorithm.
- Deployed Kafka, Zookeeper, and Neo4j on Kubernetes using Helm and Docker for a **high-availability distributed system**.

AI Game Engine, (Kotlin, Android, Minimax) Sep 2024 – Oct 2024

- Implemented **Minimax** algorithm with **Alpha-Beta Pruning** across **3** levels for an AI opponent in Kotlin Android game.
- Built a real-time game state synchronization across AI, local and Bluetooth multiplayer modes, enabling seamless turn management and win detection using reactive state management in Kotlin.

Optimizers in Deep Models, (Vision Transformers, Python, TensorFlow) Aug 2023 – Dec 2023

- Trained Vision Transformers with **7** optimizers on **CIFAR-10** and **CIFAR-100**, achieving **92.67%** and **71.26%** accuracy.
- Improved AdaBelief accuracy to **74.85%** by introducing decoupled weight decay, surpassing AdamW on **CIFAR-100**.
- Boosted **CIFAR-100** generalization using data augmentation, increasing AdamW accuracy from **68.92%** to **73.81%**.